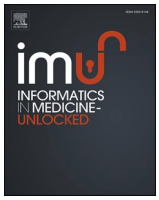
Informatics in Medicine Unlocked 48 (2024) 101518



Contents lists available at ScienceDirect

Informatics in Medicine Unlocked

journal homepage: www.elsevier.com/locate/imu





Seroprevalence of hepatitis B infection among blood donors in Western zone of Tanzania

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ARTICLE INFO

Keywords:
Seroprevalence
Hepatitis B virus
Blood donors
Western-Tanzania

ABSTRACT

Background: There is limited information on burden of hepatitis B infection in the Western zone of Tanzania. In this study, we analyzed a dataset from blood donors to determine Hepatitis B virus (HBV) seroprevalence and related socio-demographic factors among blood donors in the Western regions of Tanzania.

Material and methods: This was a cross-sectional retrospective hospital-based study, in which data were retrieved from the blood donor dataset at the Zonal Blood Transfusion Center. The analyzed information from the dataset included reported Transfusion Transmissible Infections (TTIs), which included Hepatitis B, donor demographics, donor status, donor type, donation place, and the year of donation. The analyzed data was retrieved within five years from January 2018 to December 2022. Rates of hepatitis B surface antigen (HBsAg) were determined and univariate and multivariate analyses were conducted to determine the association between infection and demographic risk factors.

Results: A total of 9604 retrospective blood donors were screened. Majority 8963 (93.3 %) were men, and most of them were under 45 years (89.6 %). Overall, HBsAg seroprevalence was 6.9 % (661), with Katavi (7.8 %) being relatively higher in the studied three regions. The highest HBsAg seroprevalence of 8.2 % (199) was found in the age group ranging from 35 to 44 years. Moreover, 2 (9.5 %) polygamists and 15 (17.1 %) car drivers had relatively high seroprevalence. Results from the multivariate analysis indicated that, car drivers (OR 5.44, 95 % CI; 2.43–12.20, $p < 0.001$), and first-time donors (OR 5.19, 95 % CI 2.56 = 10.52, $P < 0.001$), were highly associated with increased chance of getting hepatitis B infection.

Conclusion: The findings from this study indicated that, there was high seroprevalence of HBV infection in the Western regions of Tanzania during the studied time period. These findings call for more advocacy on HBV immunization for all groups of persons found at high risk for HBV infection.

1. Introduction

The World Health Organization (WHO) has declared viral hepatitis B (HBV) as a significant life-threatening global public health issue [1]. About 296 million individuals worldwide live with chronic hepatitis B, with a rate of 1.5 million new cases every year and about 0.82 million deaths annually [1]. The prevalence was found to be higher (6.2 %) in the Western Pacific Area, followed by Africa, which had 6.1 % prevalence [2]. Chronic HBV is endemic and extremely common in sub-Saharan Africa, where it affects over 8 % of the population [3], and about 12.5 % of patients receiving blood transfusions are at risk for acquiring hepatitis infection [4]. It causes liver cancer, of which, about two out of every three (2/3) incidences are brought by chronic HBV, which develops liver cirrhosis [5].

Tanzania is endemic for Hepatitis B Virus (HBV), where its prevalence was reported to be 6 % in 2018 [6]. A recent report from the National Strategic Plan for the Control of Viral Hepatitis 2018/2019 to 2022/2023, shows HBV seroprevalence from subpopulations ranging between 5.5 % and 20 % in different parts of the country [7]. Despite all government efforts for routine vaccination programs among neonates and public health care workers, the transmission rate is still high among different groups of people [2]. However, the burden of HBV among

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https://doi.org/10.1016/j.imu.2024.101518
Received 4 January 2024; Received in revised form 13 April 2024; Accepted 5 May 2024
Available online 8 May 2024
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blood donors in the Western zone of Tanzania has not been known.

The burden of Hepatitis B Virus (HBV) varies geographically among countries in endemic areas [8]. However, treatment for positively diagnosed patients is available, where antiviral therapy is effective in managing hepatitis B-infected patients [9], and the disease is preventable through vaccination, which is available as effective control measure [10] for the disease. The burden of HBV is still high in Africa, including Tanzania [3]. A study carried out in the Northern regions of Tanzania from 2017 to 2019 revealed that, the most prevalent (5.1 %) transfusion-transmitted infection (TTI) among blood donors is the hepatitis B virus (HBV) [11].

Blood transfusions and contact with bodily fluids can both result in the spread of Hepatitis B Virus, which is highly contagious [12]. The virus belongs to the hepadnaviral family, which is an enveloped DNA virus that spreads by parenteral injection and can be found in bodily fluids like blood. The virus gets into the liver through the bloodstream and reproduces in hepatocytes [13]. The risk factors for HBV infection include age, sex [14], unprotected sexual intercourse with multiple partners, tattooing, family history of hepatitis B [15], place of residence [16], occupation [17], education, alcoholism, Human Immunodeficiency Virus (HIV) [4], and blood transfusion [18]. Several studies have reported different prevalence of HBV from different zones in Tanzania across different groups [2,4,19,20] but not in the Western Zone, including its geographical distribution, and associated socio-demographic characteristics among blood donors. This study therefore aimed at generating information on seroprevalence of HBV in Western regions of Tanzania, its geographical distribution, and associated socio-demographic characteristics, to improve its management and inform policy-making process for needed appropriate interventions. The evidence generated by this research will update general knowledge on disease's current burden in Western Tanzania and its associated socio-demographic traits among study groups for appropriate intervention planning.

2. Material and Methods

2.1. Study area

This study was conducted in the Western zone of Tanzania's blood donation centers using collected retrospective data on blood donation. This study area was chosen based on the latest Demographic Health Survey (DHS) report (2022), which indicated that, a large number of women in Tabora deliver at home [21], and polygamy is most prevalent in this zone, which is one of the factors associated with Hepatitis B Virus infection.

2.2. Study design and population

This was a cross-sectional retrospective hospital-based study involving assessment retrieval and assessment of secondary data from blood donors from District hospitals for five years between January 2018 and December 2022. All collected blood was routinely tested using Chemiluminescent Microparticle Immunoassay (CMIA) technology. The study population involved all healthy male and female blood donors who were recruited from District hospitals in the Western regions. The sample size was calculated using the formula below for cross-sectional studies [22].

$$n = p(1-p)(Z/E)^2$$

Where; P= Prevalence (0.5)

Z = Level of confidence (1.96)
E = Marginal error (0.01)
n = Sample size

Recent estimates for the prevalence of HBV among blood donors in Tanzania's northern zone were 5.1 % [11]. We predicted that the Western zone of Tanzania would have a higher prevalence (10 %) among blood donors. The lack of Hepatitis B vaccination for all Tanzanian adults and challenges in getting medication for those who have been confirmed to have the infection may give credence to this hypothesis. To test this difference, we would require 9604 retrospective blood donors at a power of 100 % and an alpha set at 0.01.

Power estimation for the one-sample proportion tests

Ho: p = p0 versus Ha: p = p0

Study parameters.

alpha = 0.010
N = 9604
delta = 0.049
p0 = 0.051
pa = 0.100

The Estimated power.

power = 1.000

Where: Alpha = Significance level.

Delta = Effect size
N = Number of subjects (Sample size)
p0 = The value of the proportion under the null hypothesis, or the null proportion
pa = Alternative proportion or the value of the proportion under the alternative hypothesis
Therefore, with the sample size of n = 9604, we achieve a power of 100 %.

A total of 42550 hospital-based clean data were retrieved from the Western Zone Blood Transfusion Center database (eDelphyn) from three Regions in five years, where 9707 were from Katavi, 8267 from Kigoma, and 24576 from Tabora. The proportional calculation was conducted to obtain a total sample for each region in five years of study (total number blood donors in a region, over total number of blood donors in three regions, Times sample size required by the study), where 2190 retrospective blood donors from Katavi, 1865 from Kigoma, and 5549 from Tabora were obtained to make total of 9604 blood donors. The proportionally calculated sample size per region was evenly divided over a five-year of study period to get the number of representatives in each year. To obtain the actual representatives of 9604 sample size, all donors per year per region were randomly arranged in excel sheet and the " = RAND()" randomization formula was used to rearrange the entries using a randomization column, where the first group of donors in the list as per number of representatives required was chosen as a representative sample in that particular year.

2.3. Hepatitis B surface antigen (HBsAg) detection

All donated blood was routinely tested with Chemiluminescent Microparticle Immunoassay (CMIA) technology, which is recommended by WHO because it is very specific and ideally sensitive. It is a refined and improved version of Enzyme-Linked Immuno-Sorbent Assay (ELISA) technology. The ARCHITECT HBsAg assay is a chemiluminescent microparticle immunoassay (CMIA) for quantitative assessment of hepatitis B surface antigen (HBsAg) in human serum and plasma. It detects HBsAg using microparticles coated with monoclonal anti-HBs [19]. The blood donors were registered at blood donation centers and an assessment for eligibility criteria for blood donation was instituted using a standardized donor questionnaire. The questionnaire inquired

about participants' socio-demographic characteristics, such as age, marital status, occupation, and address, donor status (first-time donor or repeat donor), the type of donor-voluntary or replacement; and general health check of the donor in terms of diseases and risks for acquiring transmissible infections like HIV, HBV, and others. Blood donation proceeded after an individual signed a written informed consent form. According to National testing standards and procedures, every given blood was screened for TTIs, with an emphasis on HIV, HBV, Hepatitis C virus (HCV), and Syphilis-causing *Treponema pallidum* (*T. pallidum*). Candidates who obtained negative test results were marked and not recalled for the next donation. The blood that tested positive from the original screening test was discarded and put to a duplicate repeat test. Donors whose blood units tested positive after this duplicate repeat were permanently rejected, while those discovered to have medical disorders, including TTIs, were urged to seek emergency medical assistance. The seroprevalence of hepatitis B in the Western zone of Tanzania were determined from a proportion of individuals within a study population that tested positive for blood serum antibodies against hepatitis B at a particular time point.

2.4. Data management and analysis

Data from blood donors was retrieved from the Western Zone Blood Transfusion Center (WZBTC). database from the main zonal blood donation center which is located in Tabora town at Kitete Regional Hospital, from March to June 2023, and the collected variables were donation year, region, age, sex, donor status, donor type, marital status, occupation, blood group, and TTIs results.

Collected data were then cleaned, followed by data coding, and analysis using STATA version 15.1. A descriptive analysis was done to determine the infection in each year and in different groups, with different geographical areas, where the numerical variables such as age were presented as mean, median, and standard deviation. The categorical variables such as sex, age group, donor status, donor type, marital status, and occupation were summarized in the form of frequencies and percentages. The period (annual) and cumulative seroprevalence of hepatitis B infection was determined and expressed as a percentage from the proposition of all seropositive donors in the total studied population of donors. We also conducted univariate and multivariate logistic regression whereas the odds ratio (OR) were determined and corresponding 95 % (CI) confidence interval were calculated to determine factors that contribute to hepatitis B infection among blood donors and to identify statistical significance. Also, a p-value of <0.05 was considered as statistically significant threshold.

2.5. Ethical consideration

An Institutional review board from Ifakara Health Institute (IHI) granted ethical approval for this study (IHI/IRB/No: 17–2023). The permission to conduct the study using blood donor data was granted by the Ministry of Health (MoH). The information which was provided by the Western Zone Blood Transfusion Services (WZBTS) was secured under confidentiality and privacy by avoiding access to data by any unauthorized personnel and results were approved by Tanzania National Blood Transfusion Services (NBTS), National Institute for Medical Research and The Nelson Mandela African Institution of Science and Technology before publication.

3. Results

3.1. Social demographic characteristics of study participants

A total of 9604 blood donors from three regions (Katavi, Kigoma, Tabora) were involved in this study (Table 1), of which the majority (57.8 %) were from the Tabora region. The majority of blood donors recruited from this study (93.3 %) were males, and 89.6 % of all donors were younger than 45 years, 78.1 % were married and first-time blood donors accounted for 94.2 %. Additionally, 58.4 % (5608) were voluntary blood donors, while 41.6 % were replacement donors and almost half of the blood donors (48.8 %) were blood group O+(positive) as presented in Table 1.

Table 1
Socio-demographic characteristics of the study population (n = 9604).

Variable	Category	Frequency (n) (N=9604)	Percentage (%) (%=100)
Region of residence	Katavi	2190	22.8
	Kigoma	1865	19.4
	Tabora	5549	57.8
Age-	17–24 Years	2385	24.8
	25–34 Years	3805	39.6
	35–44 Years	2418	25.2
	45–54 Years	846	8.8
	55+ Years	150	1.6
Sex-	Male	8963	93.3
	Female	641	6.7
Donor status-	Repeated donation	559	5.8
	First-time donation	9045	94.2
Donor type-	Voluntary donors	5608	3996
	Replacement donors	58.4	41.6
Marital status-	Married	7500	78.1
	Not married	2047	21.3
	Polygamy	21	0.2
	Divorced	26	0.3
	Widowed	10	0.1
Blood Groups	O -Positive	4682	48.8
	O -Negative	130	1.3
	A -Positive	2559	26.6
	A -Negative	62	0.7
	B -Positive	1749	18.2
	B -Negative	33	0.3
	AB -Positive	377	3.9
	AB -Negative	12	0.1
Occupation	Students	578	6.0
	Farmer	8148	84.8
	Business	386	4.0
	Technicians	170	1.8
	Drivers	88	0.9
	Teachers	43	0.5
	Religion Leader	23	0.2
	Health Care Workers	26	0.3
	Industrial Workers	30	0.3
	Security Servants	112	1.2

The majority of participants occupations were peasants (85 %), followed by students (6 %) (Table 1).

3.2. Seroprevalence of hepatitis B

The overall seroprevalence of the hepatitis B virus in the 9604 corresponding blood donors was at 6.9 % (661/9604), of which, the hepatitis B infection was most prevalent among all tested Transfusion Transmissible Infections (TTIs) across the three regions. Other transfusion transmissible infections reported in the dataset were the HIV, Syphilis, and hepatitis C virus (Fig. 1).

The hepatitis B virus seroprevalence by HBsAg positivity was increased with age, where blood donors aged between 35 and 44 years had the highest seroprevalence (8.2 %) compared to the other age groups (Table 2). The seroprevalence of hepatitis B was slightly higher in males (6.9 %) than in females, and polygamy had a relatively higher seroprevalence of 9.5 %. However, the seroprevalence of HBV in replacement donors was 7.8 % and that of first-time donors was 7.2 %.

The seroprevalence of hepatitis B as participants in occupational activities indicated that vehicle drivers had the highest seroprevalence

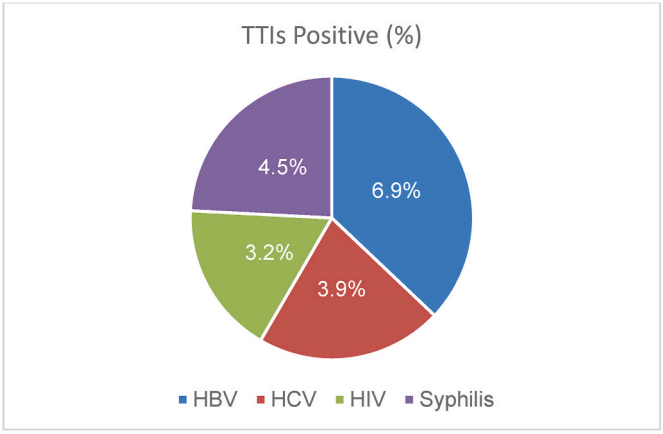


Fig. 1. Seroprevalence of Transfusion Transmissible Infections in the Western zone of Tanzania.

Table 2
Seroprevalence of HBV under different sociodemographic characteristics.

Variable	Category	Hepatitis B infection		Total
		Positive	Negative	
Region	Katavi	171(7.8 %)	2019(92.2 %)	2190
	Kigoma	131(7.0 %)	1734(92.9 %)	1865
	Tabora	359(6.5 %)	5190(92.5 %)	5549
Age	17–24 Years	121(5.1 %)	2264(94.9 %)	2385
	25–34 Years	264(6.9 %)	3541(93.1 %)	3805
	35–44 Years	199(8.2 %)	2219(91.8 %)	2418
	45–54 Years	68(8.0 %)	778(91.9 %)	846
	55+ Years	9(6.0 %)	141(94.0 %)	150
Sex	Male	622(6.9 %)	8341(93.1 %)	8963
	Female	39(6.1 %)	602(93.9 %)	641
Donor status	Repeated donation	8(1.4 %)	550(98.6 %)	558
	First-time donation	653(7.2 %)	8393(92.8 %)	9046
Donor type	Voluntary donors	350(6.2 %)	5258(93.8 %)	5608
	Replacement donors	311(7.8 %)	3685(92.2 %)	3996
Marital status	Married	575(7.7 %)	6925(92.3 %)	7500
	Not married	83(4.1 %)	1964(95.9 %)	2047
	Polygamy	2(9.5 %)	19(90.5 %)	21
	Divorced	1(3.8 %)	25(96.2 %)	26
	Widowed	0(0 %)	10(100 %)	10
Blood group	O -Positive	298(6.4 %)	4384(93.6 %)	4682
	O -Negative	9(6.9 %)	121(93.1 %)	130
	A -Positive	188(7.4 %)	2371(92.6 %)	2559
	A -Negative	2(3.2 %)	60(96.8 %)	62
	B -Positive	132(7.6 %)	1617(92.4 %)	1749
	B -Negative	2(6.1 %)	31(93.9 %)	33
	AB -Positive	29(7.7 %)	348(92.3 %)	377
	AB -Negative	1(8.3 %)	11(91.7 %)	12

of 17.1 % (Fig. 2).

Corresponding blood donors were assessed based on their blood groups and we found that, the seroprevalence of hepatitis B in blood group A+ participants was 7.35 %, while blood group B+ had a seroprevalence of 7.5 %, and blood group O+ participants had seroprevalence of 6.4 %, blood group O- had relatively higher seroprevalence of 6.9 %, blood group AB + donors had seroprevalence of 7.7 % while blood group AB- had 8.3 % and the rest as presented in Table 2.

The overall seroprevalence of Transfusion Transmissible Infections (TTIs) in the Western zone of Tanzania from 2018 to 2022 in hospital-based blood donors was 18.6 %, of which the seroprevalence of hepatitis C (HCV) in this study was 380 (3.9 %), which was slightly lower to that of Syphilis 4.5 % (431). However, HIV infection was lowest, 3.2 % (310/9604), among all the TTIs as presented in Fig. 1.

The rate of hepatitis B positivity was significantly higher in blood donors who were HIV positive (10.0 %) as per Table 3. Moreover, 380 (3.9 %) of blood donors had hepatitis C infection (Table 2), of which (17) 4.5 % were also hepatitis B positive. Furthermore, HBV infection within HCV-negative tested blood donors was significantly higher (6.9 %) compared to HCV-positive blood donors as shown in Table 3. A total of 431 (4.5 %) blood donors were infected with Syphilis causing *-Treponema pallidum,* (Table 2), of which 41 (9.5 %) were HBV and Syphilis co-infected. However, the rate of hepatitis B positive was significantly higher in blood donors who were syphilis positive compared to syphilis negative (6.8 %) (Table 3).

3.3. Periodic seroprevalence and geographical distribution of hepatitis B infection

The highest seroprevalence was observed in the Katavi region (7.8 %), followed by Kigoma (7.0 %), but lower in Tabora (6.5 %) Table 2. The year-on-year infection trend was highest in 2018 (7.6 %) but declined moderately from 2019 (6.6 %) and 2020 (6.0 %), followed by a moderate increase to 7.4 % in 2021, and 6.9 % in 2022. (Fig. 3). The seroprevalence of hepatitis B infection among blood donors in the Katavi Region was the highest (10.7 %) in the year 2018, 7.5 % in 2019, and lowest in the year 2020 (5.5 %). Moreover, the seroprevalence of HBV infection in the year 2021 and 2022 gradually increased to 7.3 % and 7.9 % respectively (Fig. 3).

Kigoma Region showed the highest seroprevalence of hepatitis B infection in the year 2020 (7.8 %), and lowest seroprevalence (6.4 %) in 2019 and 2022. But the seroprevalence in the year 2018 was 6.9 % and 7.5 % in 2021 (Fig. 3). The seroprevalence of hepatitis B virus in Tabora Region was most prevalent in the year 2021 (7.4 %) and less prevalent in 2020 (5.7 %). The seroprevalence in the year 2018 was 6.5 %, and 6.2 % in 2019, but slightly higher (6.6 %) in the year 2022 (Fig. 3).

3.4. Association between HBV infection with demographic characteristics

Univariate logistic regression analysis showed that the increased age was related to higher odds for hepatitis B infection at (OR 1.16, 95 % CI: 1.07–1.26), and this association was statistically significant ($p < 0.001$). The age group between 25 and 34 was significantly associated with hepatitis B infection (OR 1.39, 95 % CI 1.12–1.74, $p = 0.003$). A significant association with higher risk for hepatitis B infection also existed in the age group between 35 and 44 (OR 1.67, 95 % CI 1.33–2.12, $p < 0.001$) and the 45–54 years age group also showed significant association with hepatitis B infection (OR 1.63, 95 % CI 1.20–2.23, $p = 0.002$). Moreover, the first-time donors were 5.35 times more likely to be tested hepatitis B virus-positive than repeat donors at (OR 5.35, 95 % CI 2.64–10.79, $p < 0.001$). Blood donors who were married were also more likely to be hepatitis B positive than those who were single/divorced/widowed/polygamist (OR 1.96, 95 % CI 1.55–2.48, $p < 0.001$). Being a replacement donor was also associated with higher chance for being Hepatitis B positive (OR 1.26, 95 % CI 1.08–1.48, $p = 0.003$) as seen in Table 4. However, being a farmer was related to significantly increased risk for hepatitis B infection (OR 3.14, 95 % CI: 1.83–5.37, $p < 0.001$), as seen in Table 5. Businessmen were also significantly more likely to test hepatitis B positive (OR 2.67, 95 % CI; 1.36–5.23, $p = 0.004$). Being a car driver was 8.27 times more likely to be tested hepatitis B virus positive (OR 8.27, 95 % CI; 3.84–17.84, $p < 0.001$), and teachers were likewise more likely to be tested hepatitis B positive (OR 4.13, 95 % CI; 1.29–13.14, $p = 0.016$).

After adjusting for all other potential variables using a multivariate analysis, married blood donors remained at increased risk for Hepatitis B virus infection (OR 1.49, 95 % CI 1.110–2.010, $P = 0.008$). Moreover, first-time donors also remained significantly associated with increased chance of being hepatitis B positive after adjusting for other potential variables (OR 5.19, 95 % CI 2.559–10.523, $P < 0.001$) as presented in Table 4. Also, after adjusting for various confounding factors, farmers had significantly higher chance of being positive for Hepatitis B virus (OR 1.97, 95 % CI; 1.09–3.56, $p = 0.023$), and drivers remained to be strongly associated with hepatitis B infection (OR 5.44, 95 % CI;

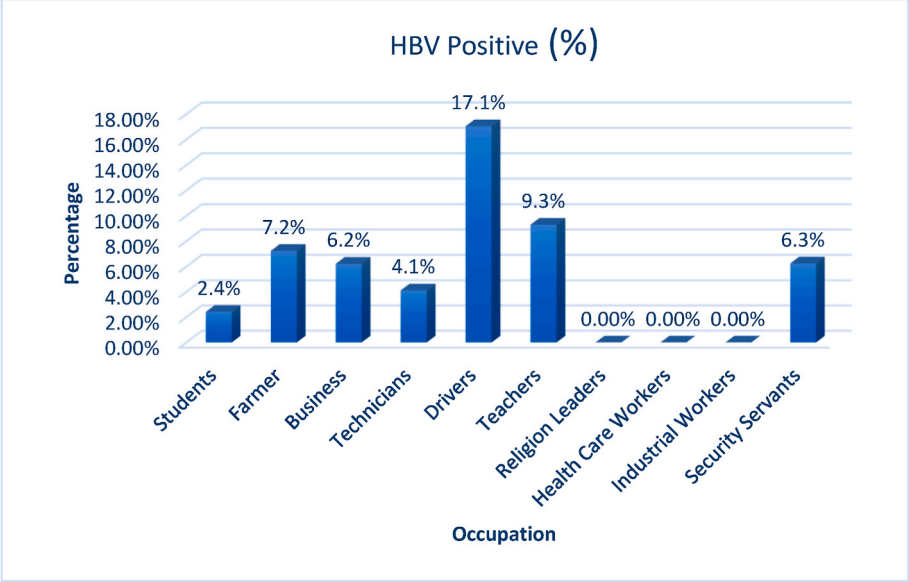


Fig. 2. Seroprevalence of HBV infection under different occupations of blood donors in the Western zone of Tanzania.

Table 3
Seroprevalence of HBV Co-infected with HCV, HIV and syphilis.

Variable	Category	Hepatitis B infection		Total
		Positive	Negative	
Hepatitis C	Hepatitis C Positive	17(4.5 %)	363(95.5 %)	380
	Hepatitis C Negative	644(7.0 %)	8580(93.0 %)	9224
HIV	HIV Positive	31(10.0 %)	279(90.0 %)	310
	HIV Negative	630(6.8 %)	8664(93.2 %)	9294
Syphilis	Syphilis Positive	41(9.5 %)	390(90.5 %)	431
	Syphilis Negative	620(6.8 %)	8553(93.2 %)	9173

2.43–12.20, $p < 0.001$), as presented in Table 5.

3.5. Association of HBV infection with other TTIs

Univariate logistic regression analysis (Table 6) revealed that, HIV-positive blood donors were significantly 1.53 times more likely to be tested hepatitis B virus positive (OR 1.53, 95 % CI; 1.04–2.23, $p = 0.029$) than HIV-negative blood donors, and also blood donors with syphilis were more likely to be hepatitis B positive than non-infected donors (OR 1.45, 95 % CI; 1.04–2.02), and were significantly associated with HBV infection. However, the HCV-infected blood donors were statistically negatively associated with HBV infection (OR 0.57, 95 % CI; 0.35–0.95, $p = 0.031$) after adjusting for other confounding variables (Table 6).

Blood donors from Tabora were 0.82 times more likely to be detected with HBV (OR 0.82, 95 % CI; 0.67 0.98) than blood donors from the Katavi region, and the association was statistically significance ($p = 0.036$) under univariate logistic regression analysis (Table 6).

4. Discussion

The viral hepatitis B infection is a significant life-threatening global public health problem, which also leads to concerns on safety of blood transfusion practices in blood transfusion centers worldwide [20]. To our knowledge, this is the first study that determined the seroprevalence of HBV infection, its geographical distribution, and associated socio-demographic characteristics among blood donors in the Western zone of Tanzania. Findings from this study are important to sensitize more up-takes of vaccinations to reduce HBV infection and the burden of this disease.

The majority of blood donors involved in this study were males, which accounted for 93.3 % of all the donors. This higher participation of males can be explained by the significant postponement of potential

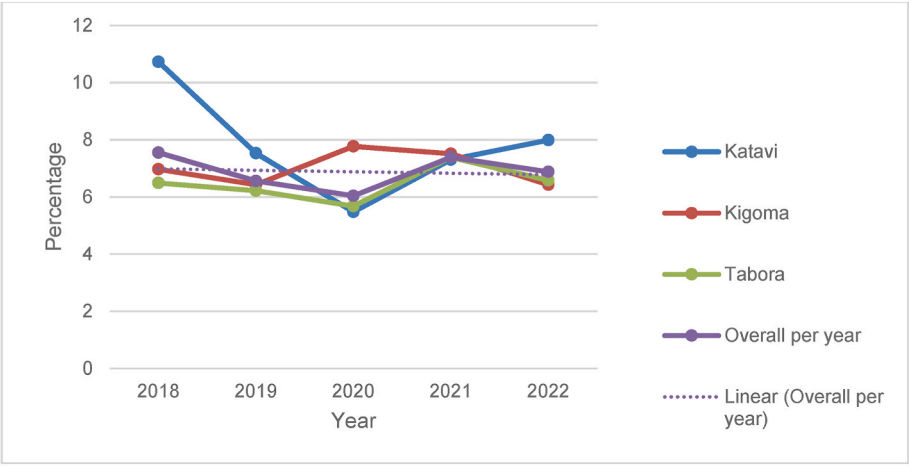


Fig. 3. Infection trend of HBV among blood donors, within a region per year, in the Western Zone of Tanzania.

Table 4
Association of hepatitis B with sociodemographic characteristics.

Variable	Category	Hepatitis B infection				
		No. of Participants	cOR (95 % CI)	p-Value	aOR(95%CI)	p-Value
Age	17–24 Years	2385	1			
	25–34 Years	3805	1.39 (1.117–1.741)	0.003	1.01 (0.781–1.311)	0.928
	35–44 Years	2418	1.67 (1.328–2.119)	<0.001	1.20 (0.912–1.591)	0.188
	45–54 Years	846	1.63 (1.201–2.225)	0.002	1.19 (0.848–1.692)	0.304
	55+ Years	150	1.19 (0.594–2.400)	0.618	0.92 (0.446–1.894)	0.821
Sex	Female	641	1			
	Male	8963	1.15 (0.824–1.607)	0.409	1.05 (0.750–1.486)	0.754
Donor status	Repeat donor	558	1			
	First-time donor	9046	5.35 (2.649–10.798)	<0.001	5.19 (2.559–10.523)	<0.001
Donor type	Voluntary donors	5608	1			
	Replacement donors	3996	1.26 (1.082–1.485)	0.003	1.21 (1.030–1.432)	0.021
Marital status	Not married	2047	1			
	Married	7500	1.96 (1.552–2.486)	<0.001	1.49 (1.110–2.010)	0.008
	Polygamy	21	2.49 (0.570–10.870)	0.225	2.06 (0.457–9.358)	0.345
	Divorced	26	0.94 (0.126–7.069)	0.957	0.68 (0.090–5.243)	0.719
	Widowed	10	0			
Blood group	AB-	12	1			
	AB+	377	0.92 (0.114–7.351)	0.935	0.88 (0.108–7.240)	0.911
	O-	130	0.82 (0.094–7.067)	0.855	0.94 (0.106–8.272)	0.955
	O+	4682	0.75 (0.096–5.810)	0.781	0.73 (0.092–5.795)	0.769
	A-	62	0.37 (0.030–4.400)	0.429	0.34 (0.028–4.231)	0.407
	A+	2559	0.87 (0.112–6.792)	0.896	0.84 (0.107–6.772)	0.882
	B-	33	0.71 (0.058–8.620)	0.788	0.67 (0.054–8.311)	0.757
	B+	1749	0.89 (0.115–7.008)	0.918	0.88 (0.110–6.992)	0.904

cOR = Crude Odds ratio, aOR = Adjusted Odds ratio.

Table 5
Association of hepatitis B with Socio-economic characteristics.

Variable	Category		Hepatitis B infection			
		No. of Participants	cOR(95 % CI)	p-Value	aOR(95%CI)	p-Value
Occupation	Student	578	1 (base)			
	Farmers	8148	3.14(1.838–5.379)	<0.001	1.97(1.096–3.569)	0.023
	Business	386	2.67(1.363–5.231)	0.004	1.81(0.884–3.711)	0.104
	Technicians	170	1.73(0.686–4.358)	0.245	1.22(0.474–3.171)	0.673
	Drivers	88	8.27(3.840–17.843)	<0.001	5.44(2.431–12.200)	<0.001
	Teachers	43	4.13(1.298–13.148)	0.016	2.56(0.778–8.455)	0.122
	Religion Leaders	23	0	–	0	–
	Health Care Workers	26	0	–	0	–
	Industrial Workers	30	0	–	0	–
	Security Servants	112	2.68(1.058–6.813)	0.038	1.67(0.639–4.412)	0.293

cOR = Crude Odds ratio, aOR = Adjusted Odds ratio.

Table 6
Association between hepatitis B and other TTIs and region of residence.

Variable	Category	Hepatitis B infection				
		No. of Participants	cOR(95 % CI)	p-Value	aOR (95%CI)	p-Value
Region of residence	Katavi	2190	1 (base)			
	Kigoma	1865	0.89(0.704–1.129)	0.343	0.93 (0.732–1.185)	0.566
	Tabora	5549	0.82(0.675–0.986)	0.036	0.83 (0.682–1.009)	0.062
TTIs	HCV	380	0.62(0.381–1.021)	0.061	0.57(0.352–0.951)	0.031*
	HIV	310	1.53(1.045–2.233)	0.029	1.44(0.980–2.114)	0.063
	Syphilis	431	1.45(1.040–2.021)	0.028	1.23(0.879–1.730)	0.224

cOR = Crude Odds ratio, aOR = Adjusted Odds ratio.

female donors due to physiological reasons, such as menstruation periods, pregnancy, lactation, and childbirth [11]. Also, the majority of blood donors (85 %) in this study were farmers (peasants) who had a significantly higher proportion among all other blood donors. Western zone regions are among the regions that have potential soil fertility which can support a variety of crops, and hence agricultural activities are the major economic activities in this study area. In our study, the majority of blood donors (89.6 %) were under the age of 45 years. This is the age group that is healthy, productive, and involved in development activities. It is therefore logical to have this finding as most African country's populations are dominated by youth. The majority of blood donors (94.2 %) were first-time donors, and voluntary non-remunerated blood donors accounted for the most (58.4 %) cases, which is higher compared with family replacement donors.

The current study has highlighted that overall seroprevalence of HBsAg among blood donors in the Western zone of Tanzania was at 6.9 %. This study indicates endemicity or previous HBV infection and hence a higher risk for HBV infection among blood donors. This seroprevalence

is within the 5.5 %–20 % range as reported by the Tanzania Ministry of Health [7] for subpopulations in Tanzania. The HBV seroprevalence in the Western regions seems to be higher than the findings from other studies, such as findings from Northern regions of Tanzania from 2017 to 2019, which reported a 5.1 % prevalence of HBV among blood donors [11], and 3.4 %, which was reported by the study conducted in Kenya east Africa involving adolescent blood donors [23], 3.8 % among mothers in Ethiopia [24], and also higher than the overall prevalence reported for East Africa (6.03 %) [25]. However, the present study findings are lower than the 11.1 % prevalence of HBV among voluntary blood donors in Côte d'Ivoire (2010–2012) [26], 11.5 % among medical students in Ethiopia [27], 7.6 % among adults in Uganda [28], and 9.8 % among blood donors in southern Ethiopia [4]. Importantly, the determined seroprevalence from this current study falls within the range of hepatitis B prevalence in Sub-Saharan Africa, which is considered high intermediate endemicity, with infection rates ranging from 5 % to 7 % [29], which agrees with [30–32] current literature that, highest HBV prevalence in the world is in Sub-Saharan Africa.

Different age groups were found to have variations in HBsAg seroprevalence, which was also reported in several previous studies globally. Moreover, the HBsAg seroprevalence was most prevalent in the 35–44 years age group (8.2 %) but low in the 17 to 24 age group and the age above 55 years. The reason for increased seroprevalence among the 35–44 age group is that this is asexually more active group, which is at a more persistent risk of exposure to HBV during their sexual lifetime, and hence their increased chance for Hepatitis B infection. Moreover, young adults are susceptible to HBV exposure risk due to their adventurous sexual behavior influenced by their age [33]. This study agrees with other studies [34] conducted in Rwanda and Côte d'Ivoire, which showed high HBV infection prevalence (5.3 %) in the age group between 35 and 44 years. Another important observation in this study was that of sex-dependent differences, which showed highest seroprevalence of HBsAg in males compared to female donors. This has also been reported in another study from Côte d'Ivoire [26]. The reason for high seroprevalence of HBsAg in males might be due to their increased exposure to occupational risks and riskier behaviors of men that involve contacts with infected body fluids.

A higher seroprevalence of HBsAg in married donors (7.7 %) may be due to the fact that, the HBV viruses are transmitted through contact with bodily fluids, including sexual intercourse, sharing of sharp objects, physical contact, and living with an infected person [33]. Married individuals may have a higher likelihood of engaging in sexual activities, and if one partner is infected, there is an increased risk of transmission within their marital relationship. The same findings on high prevalence of HBsAg in married blood donors was also reported in Yemen [35]. In our study, Farmer's and married blood donors were 1.9 and 1.5 times more likely to be tested HBV positive respectively after adjusting for other confounding factors in multivariate logistic regression, and the association was statistically significance.

The high seroprevalence of HBsAg among car drivers might be due to sexual behavior and lifestyle that leads to their increased exposure to sexually transmitted diseases risk factors. Despite having high seroprevalence, vehicle drivers were 5.4 times more likely to be tested HBV positive than other occupations after controlling for other confounding factors. These findings call for special attention in this special group during hepatitis B intervention program. The first-time donors were 5.2 times more likely to test HBV positive, and this might be because first-time donors are not aware of their health status on HBV compared to repeat donors.

In this study we found the seroprevalence of HIV-infected participants was 3.2 %. This finding agrees with other findings that, hepatitis B affects a large number of people and is more infectious than HIV [30–32]. However, hepatitis B and HIV co-infected showed a high seroprevalence of 10.0 %, which is higher than that obtained from Burkina Faso [36], and a high percentage (9.5 %) for HBV and syphilis co-infected donors were observed in our study. This indicates that, the major route of infection is through sexual intercourse and participants were exposed to unsafe sexual contacts. The hepatitis B co-infected with HIV and syphilis was statistically positively associated as revealed by univariate logistic regression, but HBV and HCV co-infected donors were negatively associated after adjusting for other confounding factors.

This study established that, HBV was the most prevalent transfusion-transmitted infection (TTI), among other TTIs determined among blood donors in this zone, where similar findings were also reported from Northern Tanzania [11]. This indicates that, the respective populations are at high risk of exposure for contracting HBV infection in this community. Nevertheless, there has been increased sensitization of HBV vaccine uptake in recent years in Tanzania, especially for the population who were born from 2003 to date, as it is being offered to infants in the routine immunization programs [7]. Yet those who succumbed to HBV infection before starting vaccinations can act as a source of infection in these communities, especially through the identified risk factors such as increased unsafe sex, living with infected persons, materials sharing, or vertical transmission. The findings from this study also signify the presence of HBV infection in the general population as seen in other studies for different groups [2,4,20] in different zones in Tanzania. All observed variations between this study and other studies could be attributed to variations in donor recruitment procedures, economic activities, geographical locations, the behavioral characteristics of different age groups, and the availability of highly sensitive screening methods [11,23,24]. Another plausible explanation is that screening blood donors for HBsAg does not eliminate the risk for HBV infection by blood transfusion, because donors with occult HBV infection who did not have detectable levels of HBsAg were tested as negative. This highlights the necessity for a more sensitive and stringent blood donor screening methodology, even in remote areas [37].

Overall seroprevalence of TTIs in the donated blood was 18.6 % within the last five years in the Western Zone of Tanzania, which is higher compared to previous study findings from Northern Tanzania [11]. Safety of blood continue to be major obstacles in Tanzania, as it is in many other sub-Saharan African nations. The high incidence of TTIs, including HBV, HCV, HIV, and syphilis, continues to be a serious problem. However, despite improvements that have been made in nations where WHO national hemo-vigilance procedures have been implemented, the danger for TTIs is still high in Tanzania, as justified by findings from this study in the Western zone of Tanzania.

We also assessed the trend of HBV infection for the consecutive five years from 2018 to 2022 and found that, year-on-year infection was not highly varied but small differences existed as in 2018 had a high overall seroprevalence of 7.6 %, and low (6.0 %) in 2020. The reason for decreased trend in HBV infections in our research population is not well clear, but it could be attributed to reduced exposure to risk factors combined with enhanced health services, such as vaccination coverage, social distancing, and hand washing, especially in the year 2020 during the COVID-19 pandemic. The decreased trend of HBV infection and increased rate of positivity for HBsAg both suggest that, horizontal rather than vertical transmission is the major source of this endemicity. However, lower seroprevalence of HBsAg among blood donors was reported in some countries [12,23]. This could be, because blood donors with clinical signs and symptoms of hepatitis have been excluded during blood donation.

The findings from this study have several significant implications for blood donation policies and public health interventions in the Western zone and Tanzania in general. Given the higher odds for hepatitis B, blood donation centers may need to implement more stringent screening measures for potential donors. This could include additional pre-testing to identify individuals with hepatitis B infection to prevent the transmission of the virus through donated blood. The findings alert for widespread vaccination campaigns against hepatitis B to reduce the overall burden of the disease. Targeted efforts can focus on populations identified as having higher odds for infection. High seroprevalence and high odds of hepatitis B in this study call for the public health campaigns

to be initiated to raise awareness about the modes of transmission, prevention measures, and the importance of getting vaccinated. This can help in reducing new infections and promoting safer behaviors. Also, based on study findings, regular screening programs can be implemented to identify and treat individuals with hepatitis B. Early detection and appropriate medical care can prevent progression of the disease and reduce the risk of transmission. A high seroprevalence may necessitate an enhancement of healthcare infrastructure to manage and treat individuals with hepatitis B effectively. This includes ensuring access to testing, vaccination, cold chain vaccine storage facilities, and medical care for those infected [38]. However, addressing the high prevalence of hepatitis B requires comprehensive and multi-faceted approach, encompassing both blood donation policies and broader public health interventions, to ensure safety blood supply while also working towards reducing the overall burden of hepatitis B in the population.

5. Conclusion and recommendations

This study involved hospital-based donated blood in the Tanzania Western Zone blood transfusion centers from 2018 to 2022, which HBV accounted for about 6.9 %. As a result, HBV remains one of the most serious hazards to blood safety in this scenario. The crucial influence of HBsAg seroprevalence included marital status, and occupation, where married participants, farmers, and car drivers are the variables, which were significantly associated with hepatitis B infection. These findings could aid not only implementation of HBV treatment options, but also in the establishment of an HBV epidemiological map in Tanzania. Improvement of blood safety in the studied regions, restricting adherence to selection and retention criteria, including voluntary, non-remunerated low-risk blood donors, are therefore highly recommended. Also, adherence to donor screening is strongly advised, with great emphasis on community-based modification of risk behaviors, individual education improvement, and assuring safe clinical practices to reduce HBV infection rates in developing countries. We highly recommend voluntary non-remunerated donors for donation centers and vaccination to all car drivers. To develop effective intervention strategies, it is also crucial to have frequent evaluations of HBV burden and trends in the Tanzanian donor community. The findings from this study emphasize increased immunization coverage against HBV to all adults at high risk, which is the most effective measure to prevent HBV infection. Compulsory screening of pregnant women is also highly recommended in endemic countries.

Several limitations were found in this study, such as the nature of the study design, which was a cross-sectional study, which does not establish causality per se. Another limitation was the nature of the study being retrospective, making it impossible for the study to provide in-depth information on risk factors for hepatitis B, which may have influenced the results in this study. Moreover, the range of risk factors evaluated in this study was limited, as it dealt with only information fed into the analyzed database, which may lead to selection bias as the data includes only hospital-based donations. Finally, systematic exclusion of individuals in blood donation and blood donors who had incomplete information in the database may bias the results because eligible blood donors may be healthier or more health-conscious, leading to under-representation of certain health conditions in the dataset. Thus, a careful interpretation of the study findings is necessary. More studies are therefore required to reveal the actual factors that contributed to high seroprevalence and high odds for hepatitis B as found in this study.

Statement and declaration

I declare that this manuscript is my original work and that it has not been submitted to any other university for the award of a degree or any other award. Also, I declare no conflict of interest regarding this study.

Ethical statement

Compliance with ethical standards

To Informatics in Medicine Unlocked- Journal.

The study titled "Seroprevalence of Hepatitis B Infection among Blood Donors in Wester zone of Tanzania" analyses data retrieved within five years from January 2018 to December 2022. The seroprevalence rates of hepatitis B surface antigen (HBsAg) were determined, and univariate and multivariate analyses were conducted to determine the association between infection and demographic risk factors.

An Institutional review board from Ifakara Health Institute (IHI) granted ethical approval for this study (IHI/IRB/No: 17–2023). The permission to conduct the study using blood donor data was granted by the Ministry of Health (MoH). The information which was provided by the Western Zone Blood Transfusion Services (WZBTS) was secured under confidentiality and privacy. The results were approved by Tanzania National Blood Transfusion Services (NBTS), the National Institute for Medical Research (Tanzania), and The Nelson Mandela African Institution of Science and Technology before publication. All procedures performed involving were in accordance with the ethical standards of the institutional and national research committee. However, for this type of study formal consent is not required. Therefore, we are kindly requesting the study to be considered for publication.

CRediT authorship contribution statement

Gadiel J. Marira: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Esther G. Kimaro:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Elingarami Sauli:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the Ifakara Health Institute for financial support and the National Blood Transfusion Services (NBTS) under the Ministry of Health-Tanzania for permitting us to conduct research by using blood donor data. Also, we thank Aidan Christopher Chaligha, Pauline Masagati Mafuru, and Gissela P. Kway for their technical support on data extraction, and Yuster L. Massanja, and Salum A. Chiamba for their contribution to data cleaning.

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